



Model-free Zigzag-Spiral search for gas source localization in uncertain environments

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ABSTRACT

This study addresses the gas source localization (GSL) problem using a mobile robot equipped with a gas sensor. Operating in dynamic and unknown environments, the robot aims to locate the source, defined as the position of the highest gas concentration. Existing model-based approaches to this maximization problem often rely on prior knowledge of gas diffusion patterns or incur high computational complexity, limiting their applicability in real-world scenarios. To address these limitations, we propose a bio-inspired, model-free optimization method, termed the Zigzag-Spiral Algorithm (ZSA), for autonomous GSL. The proposed strategy is entirely sensor-driven and requires neither predefined environmental models nor gradient information. Instead, it adaptively adjusts the robot's trajectory based on real-time gas concentration measurements. The approach consists of two stages: (1) a coarse search stage that integrates zigzag and spiral trajectories to identify the plume boundary, and (2) a fine search stage utilizing two novel spiral-based path planning strategies, namely the Inner Spiral method and the Pendular Spiral method, to accurately localize the gas source. The effectiveness of ZSA is validated through both simulations and physical experiments. Compared with the conventional Zigzag algorithm, ZSA significantly improves in localization accuracy, achieving a precision of approximately 0.1 m in real plume environments.

1. Introduction

Gas source localization (GSL) is a critical component of monitoring polluting gases [1]. The acceleration of industrialization and urbanization has increased emissions from industrial parks, municipal waste treatment plants, and other sources, exacerbating gas pollution and posing greater health risks [2]. The diffusion of gas pollution is affected by various factors, such as the sources of the gases, modes of discharge, and meteorological conditions, complicating the tracing process. Therefore, rapid and precise source localization is essential for effective monitoring of gas pollution [3].

To address this challenge, robotic olfaction using mobile platforms offers a promising solution due to its flexibility compared to static sensor networks [4,5]. However, the inherent uncertainty and turbulent nature of real-world gas dispersion. Specifically, algorithms need to be adaptable and robust to unknown environments, while achieving sufficient localization accuracy.

Bio-inspired algorithms, which mimic efficient search behaviors found in nature, provide a promising direction for GSL. Early strategies, such as the Zigzag algorithm inspired by ants [6], offer simple, reactive

search patterns [7]. As a cornerstone of bio-inspired approaches, the Zigzag algorithm has served as a widely recognized baseline due to its simplicity and generality. However, they are often limited by poor positioning accuracy [8]. Recent studies have shown that refining these biological principles can lead to performance improvements in realistic settings [9], suggesting that there is still room to enhance bio-inspired approaches.

To achieve higher positioning accuracy, more sophisticated approaches have been developed, including probabilistic methods [10] and learning-based techniques [11,12]. While these can deliver high accuracy in controlled settings, their reliance on prior knowledge, extensive data, or favorable initial conditions often limits their practical deployment [13,14]. This observation suggests an opportunity to explore whether bio-inspired methods, with their inherent simplicity and robustness, can be enhanced to achieve better localization accuracy without sacrificing their practical advantages.

Accordingly, this study explores a dual-stage approach that builds upon existing bio-inspired robotic olfaction concepts with enhancements aimed at improving positioning accuracy and adaptability in uncertain environments.

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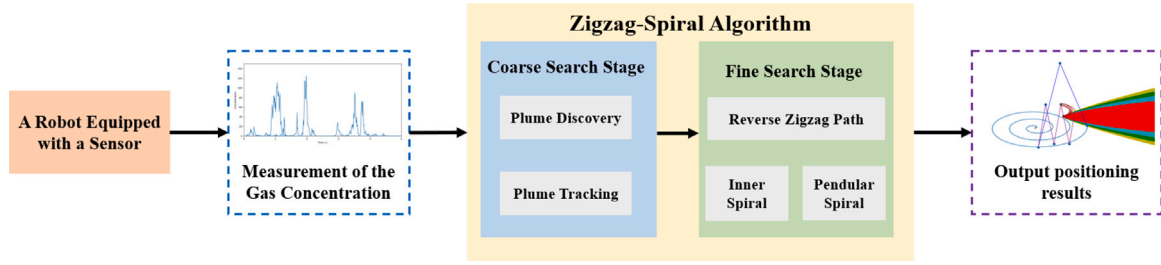


Fig. 1. Framework of the robotic olfaction system for GSL using the Zigzag-Spiral Algorithm.

The overall framework is depicted in Fig. 1. By integrating the Zigzag algorithm with the Spiral algorithm, we have developed a dual-stage bio-inspired robotic olfaction algorithm, named Zigzag-Spiral Algorithm (ZSA), to address the issues of positioning accuracy and applicability. Both real-time concentration measurements from the sensor-equipped robot and the proposed algorithm have been implemented to enable autonomous robot guidance and precise source seeking. Furthermore, we introduce a novel evaluation metric designed specifically for assessing the positioning error in GSL problems, which facilitates standardized comparisons across various experimental setups.

The main contributions of this study are summarized as follows:

(1) A model-free, bio-inspired ZSA, consisting of coarse and fine search stages, is proposed to solve the GSL problems. This algorithm relies solely on sensor concentration data without requiring prior knowledge or diffusion models.

(2) The coarse search stage guarantees that the robot discovers the plume regardless of its initial position. In the fine search stage, two spiral-based algorithms are introduced, which significantly improve positioning accuracy compared to the classical zigzag algorithm.

(3) A new criterion, η = position error/theoretical shortest path length, is introduced to evaluate GSL methods. Compared with traditional criteria such as positioning success rate [15,16], which suffer from inconsistent definitions of “success” across studies and fail to account for search complexity arising from different starting positions, the proposed metric provides a more standardized evaluation.

The remainder of this study is organized as follows: Section 2 provides a formal description of the problem, Section 3 describes the methodology, Section 4 presents the experiments, and Section 5 concludes the study.

2. Problem formulation

Consider the task of GSL within a bounded mission space $\Omega \subset \mathbb{R}^2$ in an open environment. The objective is to accurately determine the location S^* of the gas source within Ω using a mobile robot. The robot's dynamics are described as:

$$\mathbf{P}_{k+1} = \mathbf{P}_k + \mathbf{u}_k \quad (1)$$

where $\mathbf{P}_k = [P_{kx}, P_{ky}]^T \in \Omega$ denote the position of the searching robot in the current step k , $k = 0, 1, 2, \dots$, with Ω indicating the searching space, and $\mathbf{u}_k$ represents the input of the system.

We define a gas concentration function $C(\mathbf{P})$ to represent the gas concentration at a location $\mathbf{P} \in \Omega$, resulting from diffusion by the source. Equipped with a gas sensor, the robot is capable of measuring the gas concentration at its current position. In line with [17,18], we make the following assumptions:

- There is a single gas source S^* within Ω .
- Gas emissions are continuous and create a plume, indicating that the gas diffuses adequately in open air.
- The mass of the pollutant is conserved during diffusion, with no chemical transformations occurring.
- The search mission commences after the gas has sufficiently diffused.

- The wind field has a reference wind direction, and bounded disturbances are considered permissible.

Remark 1. Based on these assumptions, to ensure that the algorithm is not constrained by the model, we employ the point source Gaussian diffusion model for a single wind direction with a fixed mean wind speed, and a dynamic diffusion model for particulate matter in a time-varying wind field to simulate the values of $C(\mathbf{P}_k)$. Details of these gas models will be introduced in Section 4.

$\mathbf{P}_0$ indicates the initial position of the robot, and $\hat{S}$ represents the predicted source location. The absolute positioning error is derived as:

$$\rho = \|\hat{S} - S^*\|_2. \quad (2)$$

Considering the variability in experimental conditions, such as the environment and initial setup, we propose a new metric to quantify relative accuracy, defined as:

$$\eta = \frac{\rho}{\|\mathbf{P}_0 - S^*\|_2}. \quad (3)$$

In this study, η is introduced as a comprehensive metric to compare the performance of GSL algorithms.

3. Zigzag-spiral algorithm

In this section, the proposed ZSA with a coarse search stage and a fine search stage is described. Each search stage comprises sub-steps, with specific operational details to be systematically elaborated.

3.1. Coarse search stage

The coarse search stage, the initial phase of ZSA, is primarily designed to guide the robot into the plume and to preliminarily identify the plume's boundary. This phase consists of two steps: plume discovery and plume tracking.

3.1.1. Plume discovery

The robot is randomly deployed within the mission space Ω , which may either contain or be outside the gas plume. In cases where the robot is outside the plume, the first step involves detecting the plume. Inspired by the use of Archimedean spiral paths in search missions [19, 20], a similar strategy is employed to locate the plume.

At step k , we denote the angle of the Archimedean spiral by θ increased by $k\theta_0$. The mathematical representation of this spiral trajectory, starting from $\mathbf{P}_0 = [P_{0x}, P_{0y}]^T$, can be formulated as:

$$\begin{aligned} P_{kx} &= b\theta \cos \theta + P_{0x} \\ P_{ky} &= b\theta \sin \theta + P_{0y} \end{aligned} \quad (4)$$

where b ($b > 0$) indicates a parameter to control the distance between Spirals. The distance between Spirals is $2\pi b$. Next, the control input $\mathbf{u}_k$ is set:

$$\begin{aligned} u_{kx} &= b(k+1)\theta_0 \cos[(k+1)\theta_0] - bk\theta_0 \cos(k\theta_0) \\ u_{ky} &= b(k+1)\theta_0 \sin[(k+1)\theta_0] - bk\theta_0 \sin(k\theta_0). \end{aligned} \quad (5)$$

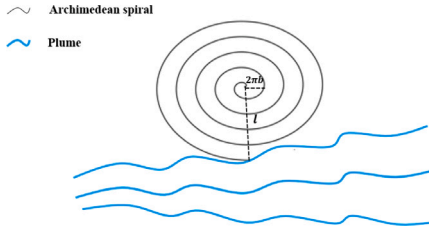
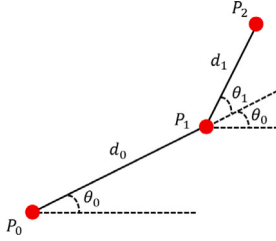


Fig. 2. Schematic of plume discovery.

Fig. 3. An example of robot motion from P_0 to P_2 .

As depicted in Fig. 2, we define the distance from the initial point to the point at which the boundary of the plume is first encountered as l . The total arc length [21] is then given by:

$$L(\theta) = \frac{b}{2} \left[\theta \sqrt{1 + \theta^2} + \ln \left(\theta + \sqrt{1 + \theta^2} \right) \right]. \quad (6)$$

Since the pole diameter $r = l = b\theta$, the arc length L decreases with the increase of the parameter b when l is fixed. This implies that increasing b can shorten the journey of the plume-searching stage. However, due to the unknown characteristics of the plume, it may either fall outside the search range or result in the plume being entirely encapsulated between spirals if b is excessively large. Therefore, this study selects a conservative value of b to ensure that the robot can enter the plume boundary. A measured value exceeding the threshold τ indicates that the robot has entered the plume.

Denote $k = s(0)$ when using the Spiral algorithm to discover the plume and begin tracking. Then, we have $C(P_{s(0)-1}) < \tau$ and $C(P_{s(0)}) \geq \tau$.

3.1.2. Plume tracking

The second step of the coarse search stage is to track the plume using the Zigzag algorithm. The primary execution strategy involves the robot advancing upstream while following a predefined angular trajectory. Upon encountering the plume area, this angle remains constant to maintain the continuity of the search process [22]. If the robot detects a low odor concentration, indicated by $C(P_k) < \tau$, the search direction will pivot at an acute angle opposite to the initial upstream trajectory. Upon re-entering the plume area, this adjusted angle is sustained to preserve the search continuity. The expression for the robot's trajectory of the Zigzag algorithm can be derived by providing the system input term u_k in (1), as follows:

$$u_k = d_k \begin{bmatrix} \cos \left(\sum_{i=s(0)}^k \theta_i \right) \\ \sin \left(\sum_{i=s(0)}^k \theta_i \right) \end{bmatrix} \quad (7)$$

where θ and d represent the rotation angle and the length of each step, respectively. The next angle is the increment of the current angle in the counterclockwise direction, implying that the robot's next movement can be designed using θ_k and d_k .

Fig. 3 shows an example of motion from P_0 to P_2 where

$$\begin{aligned} P_1 &= P_0 + d_1 [\cos \theta_0, \sin \theta_0]^T \\ P_2 &= P_1 + d_2 [\cos(\theta_0 + \theta_1), \sin(\theta_0 + \theta_1)]^T. \end{aligned} \quad (8)$$

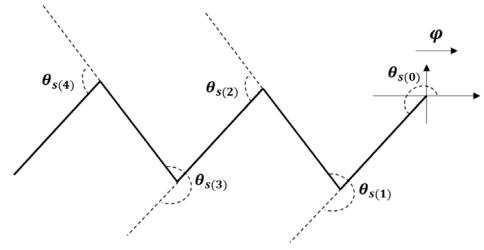


Fig. 4. The schematic diagram of Zigzag motion angle.

Furthermore, we define $\theta_{s(0)}$ as the angle relative to the reference wind direction φ : $\theta_{s(0)} = \angle(\varphi, u_{s(0)})$. To make the robot move against the wind, the rotation angles should satisfy

$$\left(\sum_{i=s(0)}^m \theta_i \right) \bmod 360^\circ \in (90^\circ, 270^\circ), \quad \forall m \in [s(0), k] \in \mathbb{N}^+. \quad (9)$$

We utilize fixed and simplified rotation angles to stabilize the search process as depicted in Fig. 4. The angles $\theta_{s(n)}$ are defined such that $\theta_{s(2j-1)} = \theta_{s(1)}$ and $\theta_{s(2j)} = \theta_{s(0)}$, where $j = 1, 2, \dots$. The subscript n serves as a marker for the turning point. Given θ_k , d_k , and other parameters, the details of the coarse search stage are described in Algorithm 1.

Algorithm 1 Coarse Search Stage

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1: Input:  $\varphi, q, \epsilon, d_1, d_2, d_3, d_s = 0, n = 1, k = 0, state = \text{INIT}$ 
2: if  $state = \text{INIT}$  and  $C(P_k) < \tau$  then
3:   Do Spiral algorithm in Plume Discovery
4: else if  $state = \text{INIT}$  and  $C(P_k) \geq \tau$  then
5:    $state \leftarrow \text{TRACK}$ 
6:    $k = s(0), \theta_k = \theta_{s(0)}$ 
7: end if
8: repeat
9:   if  $C(P_k) \geq \tau$  then
10:     $\theta_k = 0, d_k = d_1$ 
11:     $d_s \leftarrow 0$ 
12:   else if  $d_s \leq \epsilon$  then
13:     $\theta_k = 0, d_k = d_2$ 
14:     $d_s \leftarrow d_s + \|u_k\|_2$ 
15:   else
16:     $\theta_k = \theta_{s(n)}, d_k = d_3$ 
17:     $d_s \leftarrow 0$ 
18:     $n \leftarrow n + 1$ 
19:   end if
20:   Set  $u_k$  as (7)
21:    $P_{k+1} = P_k + u_k$ 
22: until  $\max(C(P_{k_1})) \leq q \times \max(C(P_{k_2})), \forall k_1 \in [s(n-1), s(n)], k_2 \in [s(n-2), s(n-1)], k_1, k_2 \in \mathbb{N}^+, n \geq 2$ 
23: Output  $P_k, C(P_k), \forall k \in \text{Coarse Search Stage}$ 

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During the coarse search stage, the plume discovery is initially conducted using the Spiral algorithm to determine the position $P_{s(0)}$. Subsequently, the Zigzag algorithm is employed for plume tracking, evaluating whether it has crossed the boundary by using the measured concentration data $C(P_k)$. Based on the value of ϵ , the robot decides whether to rotate to a larger angle to reenter the plume. This search phase continues until a specified stopping condition is met in line 22, which occurs when the maximum concentration recorded at any path point within the current line falls below or equals q times ($0 < q < 1$) the maximum concentration observed in the immediately preceding line. This criterion ensures sufficient progression through the plume. Ultimately, the coordinates of each path point are output along with the corresponding concentration value of each point. To be specific, the one with the highest concentration is the predicted source position.

Relying solely on the Zigzag algorithm may limit the accuracy of source positioning due to challenges in optimizing angles and step sizes.

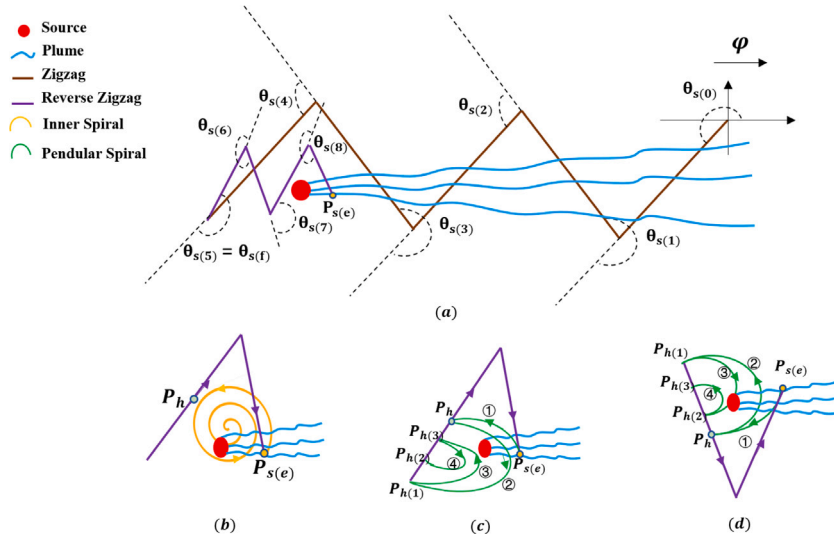


Fig. 5. Example path of the fine search stage.

To enhance precision and reduce errors, we propose the fine search stage of ZSA.

3.2. Fine search stage

We first introduce the Reverse Zigzag Path, which serves as a common component of the Inner Spiral algorithm and the Pendular Spiral algorithm. The Reverse Zigzag Path begins immediately after the end of the coarse search stage. It directs the robot to search at a finer angle along the downwind direction and re-enter the plume area, as illustrated in Fig. 5(a). Unlike the Zigzag algorithm, the specific travel path of the Reverse Zigzag Path is defined by the following equation:

$$\left(\sum_{i=s(0)}^m \theta_i \right) \bmod 360^\circ \in (-90^\circ, 90^\circ), \forall m \in [s(f), s(e)] \in \mathbb{N}^+ \quad (10)$$

and

$$\begin{aligned} \theta_{s(f+2j-1)} &= \theta_{s(f+1)} \\ \theta_{s(f+2j)} &= \theta_{s(f+2)} \end{aligned} \quad (11)$$

where $s(f)$ and $s(e)$ represent the starting and ending indices of the Reverse Zigzag Path, respectively. The selection of step sizes changes from d_1, d_2 , and d_3 to d_4, d_5 , and d_6 , respectively. The turning condition is defined as $d_s > \epsilon'$. The path terminates when the robot moves away from the plume boundary, such that $C(P_{s(e-1)}) \geq \tau$ and $C(P_{s(e)}) < \tau$.

3.2.1. Inner spiral algorithm

The Inner Spiral algorithm utilizes data gathered from the previously explored field to gradually reduce the polar diameter of the robot's spiral movement. At the current position of the robot, $P_{s(e)}$, as shown in Fig. 5(b), and assuming that the midpoint of the previous line in the Reverse Zigzag Path is P_h , the robot initiates an inner spiral motion starting with a pole diameter r_0 :

$$r_0 = \frac{\|P_{s(e)} - P_h\|_2}{2} \quad (12)$$

It then traverses a smaller area. Given a small positive constant δ , the polar radius is set as $r = r_0 - b_1 \Theta$ and $r \geq \delta$. When the polar radius falls below this threshold, the robot ceases movement and returns the predicted source location.

For a counterclockwise inner spiral starting from the origin, the coordinate points $[P'_{kx}, P'_{ky}]^T$ are derived by (4) as follows:

$$\begin{bmatrix} P'_{kx} \\ P'_{ky} \end{bmatrix} = \begin{bmatrix} r \cos \Theta - r_0 \\ r \sin \Theta \end{bmatrix} \quad (13)$$

where $r = r_0 - b_1 \Theta$, and b_1 is a constant. The rotation matrix is defined as:

$$R(\Phi) = \begin{bmatrix} \cos \Phi & -\sin \Phi \\ \sin \Phi & \cos \Phi \end{bmatrix} \quad (14)$$

where Φ indicates the rotation angle. The coordinates $[P_{kx}, P_{ky}]^T$ of the Inner Spiral trajectory are then set by the rotation matrix and translated to the point $[P_{s(e)x}, P_{s(e)y}]^T$:

$$\begin{aligned} \begin{bmatrix} P_{kx} \\ P_{ky} \end{bmatrix} &= \begin{bmatrix} \cos \Phi & -\sin \Phi \\ \sin \Phi & \cos \Phi \end{bmatrix} \begin{bmatrix} P'_{kx} \\ P'_{ky} \end{bmatrix} + \begin{bmatrix} P_{s(e)x} \\ P_{s(e)y} \end{bmatrix} \\ &= \begin{bmatrix} P_{s(e)x} + r \cos(\Phi + \Theta) - r_0 \cos \Phi \\ P_{s(e)y} + r \sin(\Phi + \Theta) - r_0 \sin \Phi \end{bmatrix} \end{aligned} \quad (15)$$

where

$$\begin{aligned} \Phi &= \arctan \frac{P_{hy} - P_{s(e)y}}{P_{hx} - P_{s(e)x}} \\ \Theta &= (k - s(e))\Theta' \\ k &\geq s(e), k \in \mathbb{N}^+. \end{aligned} \quad (16)$$

3.2.2. Pendular spiral algorithm

The path planning of the Inner Spiral algorithm utilizes only the position of point P_h in the Reverse Zigzag Path, which limits its traversal range. To fully utilize the penultimate line in the Reverse Zigzag Path, we propose the Pendular Spiral algorithm. This approach can be described as a process of gradually approaching the source position in the upwind direction through Pendular Spiral motion.

The following description corresponds to Fig. 5(c) of the algorithm: when the robot is positioned at the lower boundary of the plume, it selects the lower section of the straight line, which is divided into γ equal parts (with $\gamma = 3$ in the figure). The division points are sequentially labeled along the travel path as $P_{h(1)}, P_{h(2)}, P_{h(3)}$ and $P_{h(4)} = P_{h(\gamma+1)} = P_h$. The robot follows a specific sequence of paths: $P_{s(e)} \rightarrow P_h \rightarrow P_{h(1)} \rightarrow P_{h(2)} \rightarrow P_{h(3)} \rightarrow P_{h(4)} \rightarrow P_{h(3)} \rightarrow P_{h(2)} \rightarrow P_{h(1)} \rightarrow P_h$. Along these paths, its trajectory alternates between counterclockwise and clockwise spirals. If the robot is positioned at the upper boundary, it executes a Pendular Spiral movement between the upper endpoint and the midpoint, as illustrated in Fig. 5(d).

In accordance with (4), a spiral motion is implemented between each pair of determined points (assumed to be from P_i to P_j):

$$\begin{aligned} P_{jx} &= r \cos \Theta + P_{ix} \\ P_{jy} &= r \sin \Theta + P_{iy} \end{aligned} \quad (17)$$

where θ denotes the polar angle. The endpoint in polar coordinates is determined by $r_j = \sqrt{(P_{jx} - P_{ix})^2 + (P_{jy} - P_{iy})^2}$ and $\theta_j = \arctan(\frac{P_{jy} - P_{iy}}{P_{jx} - P_{ix}})$. The starting angle is given by: $\theta_i = \theta_j + a \cdot \frac{r_j}{b_2}$ where b_2 indicates the expansion coefficient, and a represents a direction factor defined as:

$$a = \begin{cases} 1, & \text{even-spiral} \\ -1, & \text{odd-spiral} \end{cases} \quad (18)$$

The spiral is generated by varying θ from θ_i to θ_j with a corresponding radius.

$$r(\theta) = b_2 \cdot (\theta_i - \theta) \cdot a. \quad (19)$$

When it is detected that the concentration of all path points on a certain spiral segment is lower than the threshold τ , the robot is considered to have exited the plume boundary. At this point, the robot stops, and the position of the point with maximum concentration is predicted to be the source position.

Both the Inner Spiral and Pendulum Spiral algorithms utilize a spiral pattern for translational and rotational movements, owing to their efficient and straightforward traversal mechanisms. The spiral path offers a design that is simple yet effective, with minimal computational requirements.

3.3. Parameter settings

The parameter configuration of the proposed Zigzag-Spiral Algorithm (ZSA) is fundamentally linked to the dimensions of the search region. This design ensures that the algorithm's operational scale is appropriate for the environment in which localization is performed. In this section, we present the specific parameter settings.

It is important to note that the parameter ranges provided below are recommended values based on geometric considerations and experimental results within our study. Since the ZSA operates primarily on a two-dimensional geometric plane and is designed as a model-free strategy, the selection of these parameters is not derived from rigorous mathematical proofs. Therefore, these parameters are provided as approximate ranges rather than as specific, fixed values. All simulations and experiments presented in subsequent sections utilize parameters selected according to the following guidelines.

Assuming the search area can be approximated as a planar rectangle, the length of its shorter side is defined as c . For the zigzag trajectory, the parameters to be considered are d_1 to d_3 for the coarse search phase, and d_4 to d_6 for the fine search phase:

- **Coarse Search Basic Step Length (d_1):** The fundamental stride length during plume-tracking zigzag motion in the coarse search phase. Its value is set to $0.1\sqrt{c} \leq d_1 \leq 0.3\sqrt{c}$ to ensure adequate cross-plume sampling while maintaining smooth robot navigation.
- **Plume-Exit Verification Step Length (d_2):** Used to verify whether the robot has completely exited the plume. Its value is set to $0.5d_1$, smaller than d_1 , to enable precise determination of the plume boundary.
- **Plume Re-entry Step Length (d_3):** The distance traveled after confirming plume exit to ensure the robot returns to within the plume. It is set to a relatively large value of $8d_1 \leq d_3 \leq 10d_1$.
- **Fine Search Basic Step Length (d_4):** The fundamental stride length used during the fine search phase. To achieve higher spatial resolution, it is set much smaller than d_1 , specifically $0.2d_1$.
- **Fine Search Plume-Exit Verification Step Length (d_5):** Analogous to d_2 but for the fine search phase. Its value is set to half of d_4 , like $0.1d_1$, for precise boundary detection.
- **Fine Search Plume Re-entry Step Length (d_6):** Analogous to d_3 but for the fine search phase. It is set to $10d_4$, which equals $2d_1$, to ensure re-entry into the plume during high-resolution localization.

The setting of spiral parameters mainly includes the spiral coefficient b for the coarse search phase, the spiral coefficient b_1 for the Inner Spiral algorithm, and the spiral coefficient b_2 for the Pendulum Spiral algorithm:

- **Outer Spiral Coefficient (b):** This coefficient defines the turn spacing of the Archimedean spiral during the coarse search phase. Assuming n spiral turns are required to cover approximately half of the search width c , the relationship $2\pi bn = \frac{c}{2}$ yields $b = \frac{c}{4\pi n}$. To ensure sufficient coverage, it is recommended that $n \geq 3$, which corresponds to $0 < b \leq \frac{c}{12\pi}$.
- **Inner Spiral Coefficient (b_1):** Defines the turn spacing for the fine-search inner spiral. To achieve higher resolution, it is set smaller than b , specifically $0.1b \leq b_1 \leq 0.2b$.
- **Pendulum Spiral Coefficient (b_2):** Defines the variable turn spacing for the pendulum spiral algorithm. It is designed as a slowly increasing function of the segment index i to adapt the search scale, given by $b_2 = \frac{c}{20} \cdot \log(i + 1)$.

In addition, other parameters require defined criteria for their configuration, including the concentration threshold τ , the path length thresholds ϵ and ϵ' for determining turn-back conditions in zigzag and reverse zigzag paths, as well as the stopping parameter q for the coarse search phase:

- **Concentration threshold (τ):** In simulation experiments, τ is determined based on the concentration range computed from the known dispersion model. In practice, it should be set according to the sensor's response characteristics, which will be discussed in detail in Section 4.
- **Turn-back thresholds (ϵ and ϵ'):** These parameters define the path length thresholds for triggering a turn-back in zigzag and reverse zigzag paths, respectively. A typical setting is $3d_2 \leq \epsilon \leq 5d_2$ and $3d_4 \leq \epsilon' \leq 5d_4$.
- **Coarse search stopping parameter (q):** This parameter controls the termination condition of the coarse search phase. It is generally chosen in the range $q \in [0.1, 0.2]$.

4. Simulation experiment

We carried out simulation experiments in a Gaussian diffusion model with a fixed wind field and a filament-based model under a dynamic wind field for validating the effectiveness of the proposed algorithm.

4.1. Simulation in Gaussian diffusion model

The Gaussian diffusion model describes the diffusion of emissions from a continuous point source in an unbounded domain and follows widely accepted standards for simulating airborne pollutant transport [17,23]. In this study, the source location S^* is positioned at the origin. Assuming a wind direction defined as $\psi = \vec{OX}$, we formulate the gas concentration distribution based on the assumptions outlined in Section 2. Under these conditions, the concentration at any sampling point is defined as:

$$C(x, y) = \frac{ER}{2\pi\bar{u}\sigma_y\sigma_z} e^{-\frac{y^2}{2\sigma_y^2}} \quad (20)$$

where σ_y indicates the horizontal diffusion parameter, and σ_z denotes the vertical diffusion parameter. $\bar{u}$ represents the average wind speed, and ER denotes the emission rate.

In this study, the variables σ_y and σ_z in the model are fitted using the following equations: $\sigma_y = \gamma_1 x^{\alpha_1}$, $\sigma_z = \gamma_2 x^{\alpha_2}$. The fitting parameters for specific wind class conditions are obtained through experimentation. The specific data and model plots from the experiments are presented in Table 1 and Fig. 6, respectively.

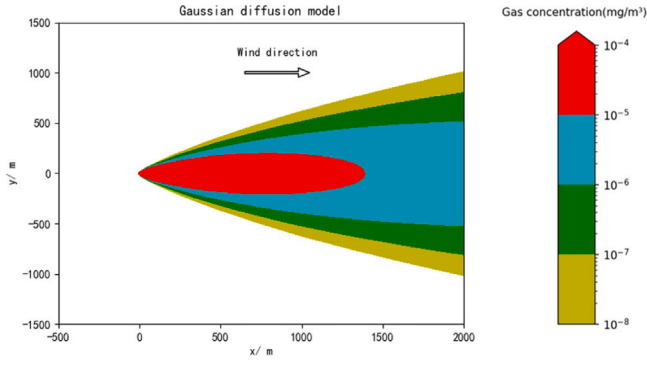


Fig. 6. Gaussian diffusion model. For the experiment in Section 4, environmental parameters are known: the wind direction is $\psi = \overrightarrow{OX}$, average wind speed $\bar{u} = 1$ m/s, emission rate $ER = 1$ mg/s. The area of interest is expressed as: $\Omega = \{(x, y) \mid -500 \leq x \leq 2000, -1500 \leq y \leq 1500\}$.

Table 1
Parameters for the calculation of σ .

Distance(m)	γ_1	α_1	γ_2	α_2
$0 < x \leq 500$	0.281846	0.91437	0.12719	0.96144
$500 < x \leq 1000$	0.281846	0.91437	0.057025	1.09356
$1000 < x < 10000$	0.396353	0.86501	0.057025	1.09356

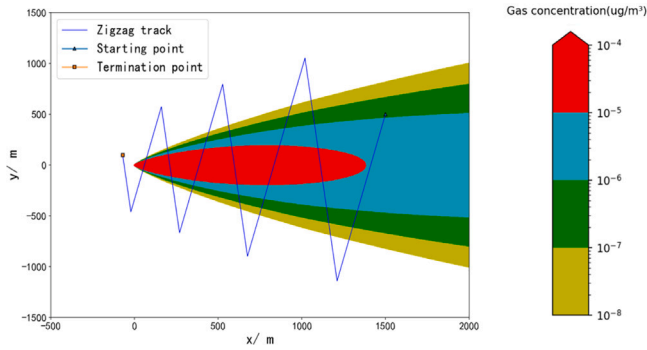


Fig. 7. Simulation results of the Zigzag algorithm. The triangle represents the starting point A, and the square represents the ending point. The blue line denotes the path of the robot.

According to the gas diffusion model, the calculated gas concentration is categorized into four regions ranging from 10^{-8} mg/m³ to 10^{-4} mg/m³, each represented by different colors. Concentrations lower than 10^{-8} mg/m³ are disregarded, setting $\tau = 10^{-8}$ mg/m³. This approach is inspired by [24], which models both σ_y and σ_z as linear functions of x , and by [25], which models σ_y using an empirical power law function of x and σ_z using a Nike function of x to minimize errors.

4.1.1. Downwind plume exploration

Three different initial positions were selected, including $A = [1500, 500]^T$, $B = [1500, 1000]^T$, and $C = [1000, -1000]^T$, for evaluating the search performance of the proposed ZSA. The first position is inside the plume, and the other two are outside, facilitating the comparison of experimental results. Table 2 presents the Zigzag Path parameters, including the angle of the turning point $\theta_{s(n)}$, step length d , and step count configurations for the mobile robot.

Simulation experiments were initially conducted using the Zigzag algorithm. Fig. 7 presents the simulation results starting from point A, where the robot gradually converges toward the vicinity of the source. The termination criterion for the search is set to $q = 0.2$, as detailed in Section 3. This criterion ensures that the robot ceases movement after

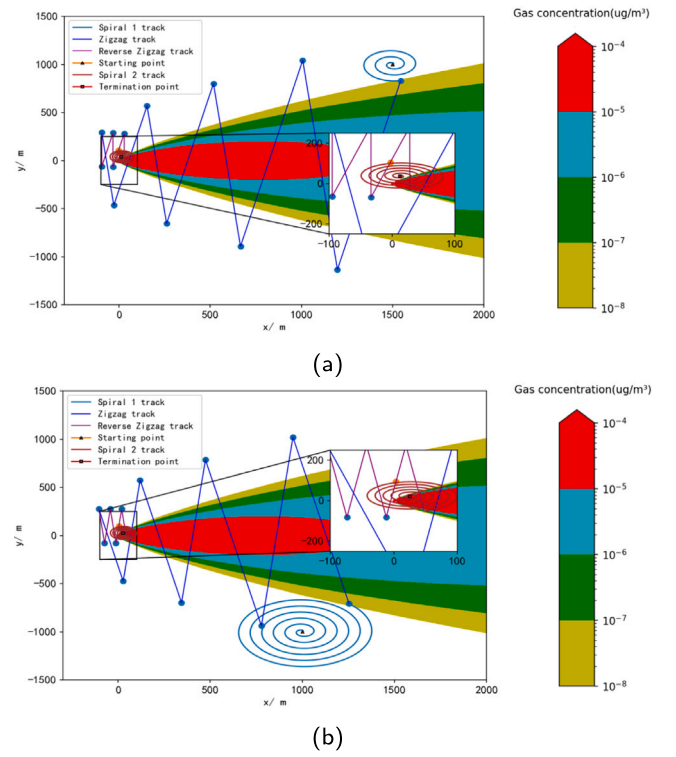


Fig. 8. Simulation results of the Inner Spiral algorithm. Figures (a) and (b) display the results for starting points B and C, respectively. The blue line segment represents the Zigzag path, the purple line segment indicates the Reverse Zigzag path, and the brown line segment represents the Inner Spiral trajectory.

exiting the plume and subsequently moves in the upwind direction, underscoring the importance of sufficiently traversing the plume area for a comprehensive estimation of the source location.

For the ZSA component, simulation experiments were conducted from three starting positions as previously mentioned. Figs. 8 and 9 illustrate the outcomes of the simulations for the Inner Spiral algorithm and the Pendular Spiral algorithm, respectively. During the coarse search stage (see Section 3.1), the spiral parameters set for plume discovery are $b = 10$ and $\theta_0 = 0.01\pi$. In the fine search stage, the parameters for the Inner Spiral algorithm are $b_1 = 2$ and $\theta'_0 = 0.005\pi$. Conversely, the Pendular Spiral algorithm uses $b_2 = 150 \log(i+1)$ (where i represents the line segment number, as depicted in Fig. 5) and $\gamma = 10$.

The settings of these parameters are derived from the guidelines established in Section 3.3, which are based on the relationship with the shorter side length of the search area. For the Gaussian diffusion model selected in this paper, the length of the shorter side of the exploration area c is 3000 m. Therefore, the basic step length d_1 lies in the interval: $d_1 \in [0.1\sqrt{3000}, 0.3\sqrt{3000}] \approx [5.48, 16.43]$. The basic spiral coefficient b satisfies $b = \frac{c}{4\pi n}$ with $n \geq 3$. For $c = 3000$, this gives $0 < b \leq 79.58$.

Other parameters can be obtained by referring to the proportional relationships given in Section 3.3. Since the proposed ZSA in this paper is developed based on geometric plane traversal relationships and real-time feedback from concentration measurements, rather than on rigorous mathematical proofs, parameter values (such as d_2 and d_3) are allowed to have a certain degree of deviation during selection.

Using the evaluation indicator η , introduced in Section 2, a quantitative analysis was carried out on these three methods. As presented in Table 3, both the Inner Spiral and Pendular Spiral algorithms achieve lower values of η , demonstrating enhanced search accuracy. Additionally, 100 random experiments were conducted. The box plot in Fig. 10 shows the error distributions for the three algorithms. The

Table 2
Parameters in Zigzag search.

Path type	Starting point	Parameter $\theta_{s(n)}$	Parameter d
Zigzag	A,B	$\theta_{s(0)} = 260^\circ, \theta_{s(1)} = 195^\circ, \theta_{s(2)} = 165^\circ$	$d_1, d_2, d_3 = 15 \text{ m}, 10 \text{ m}, 150 \text{ m}$
	C	$\theta_{s(0)} = 100^\circ, \theta_{s(1)} = 165^\circ, \theta_{s(2)} = 195^\circ$	
Reverse Zigzag	A,B	$\theta_{s(f)} = 175^\circ, \theta_{s(f+1)} = 190^\circ, \theta_{s(f+2)} = 160^\circ$	$d_4, d_5, d_6 = 1 \text{ m}, 5 \text{ m}, 50 \text{ m}$
	C	$\theta_{s(f)} = 175^\circ, \theta_{s(f+1)} = 165^\circ, \theta_{s(f+2)} = 195^\circ$	

Table 3
Downwind plume experimental results.

Method	Starting point	Predicted source location	Error(m)	$\eta(10^{-3})$
Zigzag algorithm	A	(60.05, 2.17)	60.09	38.01
	B	(20.18, -7.17)	21.41	12.87
	C	(52.21, 0.63)	52.21	36.92
ZSA with Inner Spiral	A	(6.58, 0.52)	6.60	4.17
	B	(10.16, 3.22)	10.66	5.92
	C	(8.25, 1.16)	8.34	5.89
ZSA with Pendular Spiral	A	(0.34, 0.11)	0.36	0.22
	B	(5.44, 1.27)	5.59	3.10
	C	(7.88, 0.11)	7.88	5.58

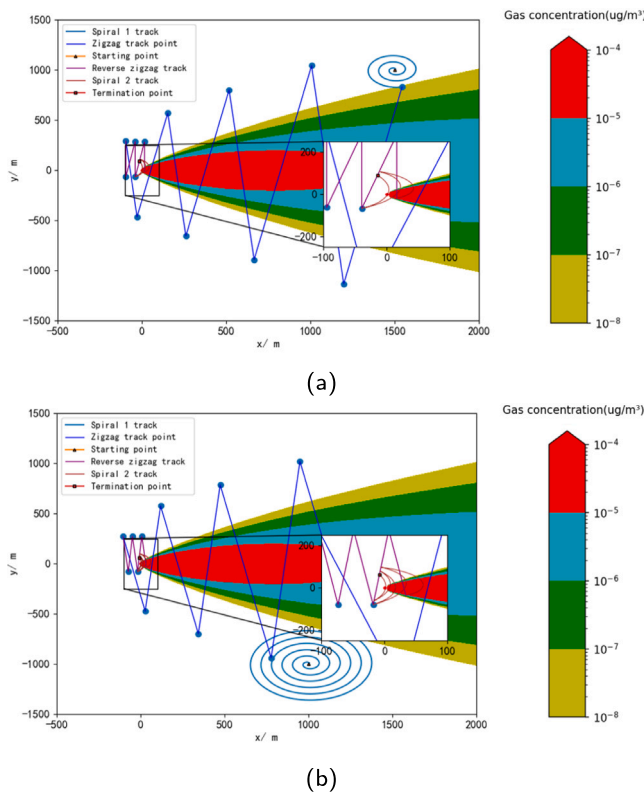


Fig. 9. Simulation results of the Pendular Spiral algorithm. Figures (a) and (b) present the results for starting points B and C, respectively. The Spiral 2 track in the figure is replaced by the Pendular Spiral trajectory. The rest is the same as in Fig. 8.

results indicate significant enhancements in positioning accuracy for the proposed algorithms, with the Pendular Spiral algorithm exhibiting a slight edge. ZSA not only improves search accuracy but also integrates spiral trajectories for plume discovery, increasing its versatility in addressing source localization problems. The subsequent section will discuss scenarios where the robot's starting point is positioned behind the source, validating the applicability of the proposed algorithm.

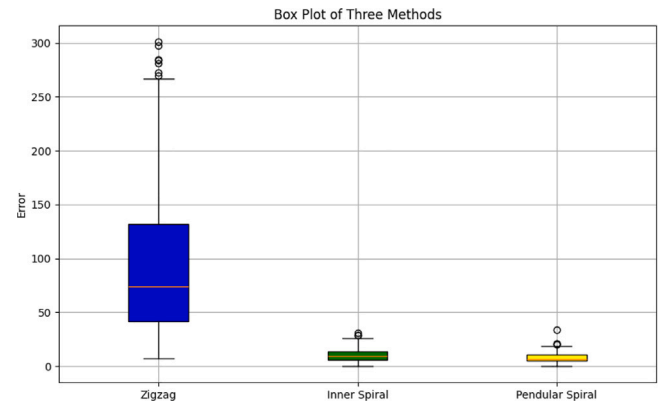


Fig. 10. This box plot compares the error distributions of the three methods. The boxes represent the interquartile range (IQR), while the horizontal line inside each box indicates the median error.

4.1.2. Search starting at the back of the source

Due to the uncertainty of environmental factors such as wind speed and direction, the distribution of gas concentration does not solely occur downwind but may also appear in any area near the source. Therefore, in addition to accuracy, the proposed algorithm must include an effective strategy to access the plume from arbitrary starting positions within the search area, thereby meeting the requirements for algorithmic flexibility.

The classical Zigzag algorithm shows limitations, especially in scenarios where the robot's starting position is behind the source. In contrast, the proposed strategy effectively overcomes this critical challenge. Within the ZSA framework, the spiral trajectory used during the coarse search stage offers significant advantages. As shown in Fig. 11, the starting point in the experiment is set at $D = (-100, -100)$. The corresponding experimental results are summarized in Table 4. Although the robot covers a greater distance, due to the extended spiral approach and less-than-optimal parameter settings, compared to the downwind scenario discussed in Section 4.1.1, the algorithm's applicability is clearly demonstrated. Overall, the proposed approach is effective in such nonstandard environments and achieves improved localization performance.

In terms of both applicability and positioning accuracy, ZSA demonstrates greater feasibility than the Zigzag algorithm, as evidenced by the experimental results. Notably, ZSA requires minimal prior knowledge and remains robust under environmental variability. The following

Table 4
Experimental results with search starting from the back of the source (starting point D).

Method	Predicted source location	Error(m)	$\eta(10^{-3})$
Zigzag algorithm	N/A		
ZSA with Inner Spiral	(15.07, 0.28)	15.07	0.107
ZSA with Pendular Spiral	(5.88, -0.17)	5.89	0.041

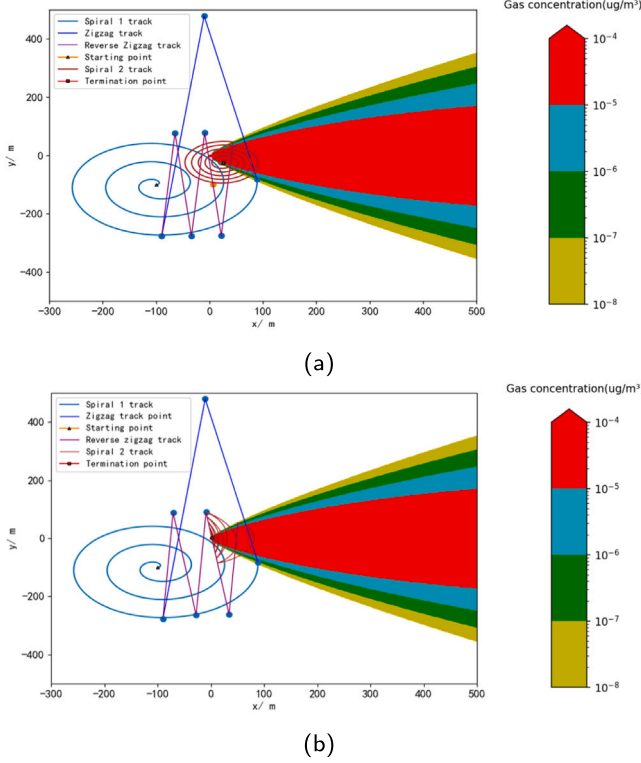


Fig. 11. Simulation result using the Inner Spiral algorithm and the Pendular Spiral algorithm with starting point D.

section presents experiments conducted in dynamic environments to further demonstrate this robustness.

4.2. Simulation in filament-based model

In this section, we validate the robustness and applicability of the proposed algorithm within a dynamic, turbulent environment. We employ an experimental setup based on the filament-based odor plume model described by Farrell et al. [26]. This model offers a computationally efficient means to simulate the diffusion of odor plumes in turbulent flows, capturing essential characteristics such as short-term intermittency and diffusive effects. It represents odor plumes as sequences of puffs comprising multiple filaments, which are transported and dispersed by a dynamic wind field.

The wind field is modeled using a set of partial differential equations that describe the advection and diffusion of wind velocity components. The advection equations for the wind velocity components v_{ax} and v_{ay} are formulated into:

$$\begin{aligned} \frac{\partial v_{ax}}{\partial t} = & -v_{ax} \frac{\partial v_{ax}}{\partial x} - v_{ay} \frac{\partial v_{ax}}{\partial y} \\ & + \frac{1}{2} K_x \frac{\partial^2 v_{ax}}{\partial x^2} + \frac{1}{2} K_y \frac{\partial^2 v_{ax}}{\partial y^2} \end{aligned} \quad (21)$$

$$\begin{aligned} \frac{\partial v_{ay}}{\partial t} = & -v_{ax} \frac{\partial v_{ay}}{\partial x} - v_{ay} \frac{\partial v_{ay}}{\partial y} \\ & + \frac{1}{2} K_x \frac{\partial^2 v_{ay}}{\partial x^2} + \frac{1}{2} K_y \frac{\partial^2 v_{ay}}{\partial y^2} \end{aligned} \quad (22)$$

Table 5
Wind model parameters.

Parameter	Value
Mean x-component of wind velocity u_{av}	2.0 m/s
Mean y-component of wind velocity v_{av}	0.0 m/s
Diffusivity constant in x direction K_x	10.0 m ² /s
Diffusivity constant in y direction K_y	10.0 m ² /s
noise grain G	5.0
noise damp $\frac{b}{2\sqrt{a}}$	0.1
noise bandwidth $\sqrt{a}$	0.2

Table 6
Plume model parameters.

Parameter	Value
central rel diff scale σ_v	2.0 m/ $\sqrt{s}$
puff initial radius $R(0)$	0.001 ^{0.5} m
puff diffusion rate γ	0.001 m ² /s
puff molecular amount Q	8.3×10^8 molecules

In these equations, K_x and K_y denote the diffusivity coefficients in the x and y directions, respectively. They describe the temporal and spatial variations in wind velocity, influenced by advection and diffusion processes. Each filament within the plume is characterized by its center position p_i and radius R_i . The movement of the filament's center is governed by the wind velocity field:

$$\dot{p}_i = v_a(p_i) + v_m(p_i) \quad (23)$$

Here, $v_a(p_i)$ represents the advective wind velocity at the filament's position, and $v_m(p_i)$ denotes the mesoscale wind velocity component. The radius of each filament expands over time according to:

$$\dot{R}_i = \gamma R_i(0)^{1/3} \quad (24)$$

where γ denotes a parameter that controls the growth rate of the filament. This growth process mimics the diffusion of odor molecules within the filament over time. The odor concentration at a specific position z and time t is calculated as the aggregate of contributions from all filaments:

$$C(z, t) = \sum_{i=1}^N C_i(z, t). \quad (25)$$

The concentration contribution from each filament follows a Gaussian distribution:

$$C_i(z, t) = \frac{Q}{R_i^3(t) \sqrt{8\pi^3}} \exp\left(-\frac{\|z - p_i(t)\|^2}{2R_i^2(t)}\right) \quad (26)$$

where Q indicates the amount of odor molecules per filament, and $R_i(t)$ represents the radius of the i th filament at time t . This model ensures that the odor concentration peaks near the filament's center and decreases with distance from the center.

The experiments were executed using the Pompy package, and the source code is accessible at the Pompy GitHub repository (<https://github.com/INSERTRobotics/pompy>). The Pompy package utilizes varied colored noise model parameters, as listed in Tables 5 and 6. Fig. 12 illustrates the dynamic plume operating from 2 s to 30 s.

In alignment with the experiments conducted under the Gaussian model in 4.1, further experiments were carried out to evaluate the GSL performance of three methods under conditions of downwind-start seeking and source-backside seeking. Considering the dynamic

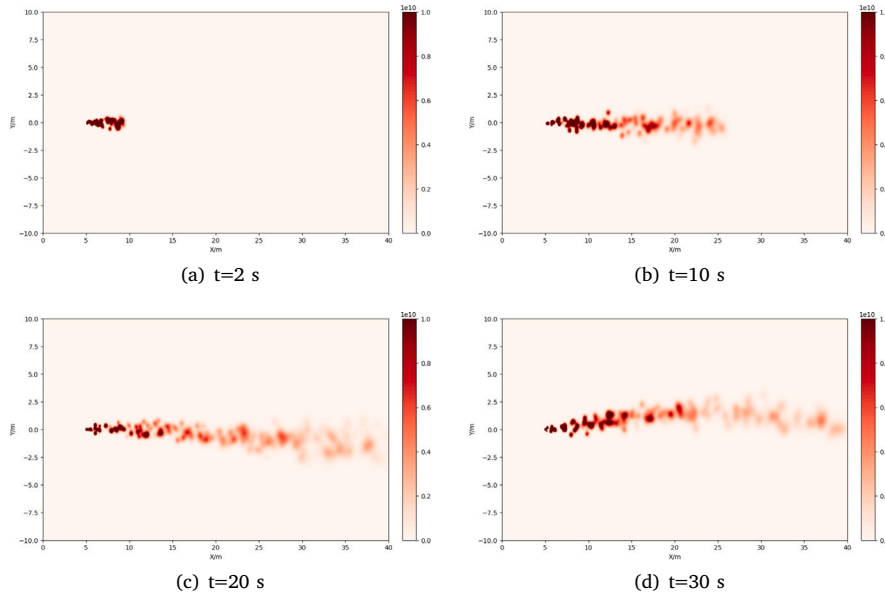


Fig. 12. Schematic of the diffusion process in filament-based dynamic plume modeling from 2 s to 30 s. Concentration values range from 0 to 10^{10} . The plume's extent is indicated by the red area, with deeper reds representing higher concentration values.

Table 7

Parameters in Zigzag Search (dynamic plume model).

Path type	Starting point	Parameter $\theta_{s(n)}$	Parameter d
Zigzag	E	$\theta_{s(0)} = 245^\circ, \theta_{s(1)} = 230^\circ, \theta_{s(2)} = 130^\circ$	$d_1, d_2, d_3 = 0.5 \text{ m}, 0.2 \text{ m}, 4\text{m}$
	F	$\theta_{s(0)} = 115^\circ, \theta_{s(1)} = 130^\circ, \theta_{s(2)} = 230^\circ$	
Reverse Zigzag	E	$\theta_{s(f)} = 230^\circ, \theta_{s(f+1)} = 190^\circ, \theta_{s(f+2)} = 170^\circ$	$d_4, d_5, d_6 = 0.1 \text{ m}, 0.1 \text{ m}, 2\text{m}$
	F	$\theta_{s(f)} = 130^\circ, \theta_{s(f+1)} = 170^\circ, \theta_{s(f+2)} = 190^\circ$	

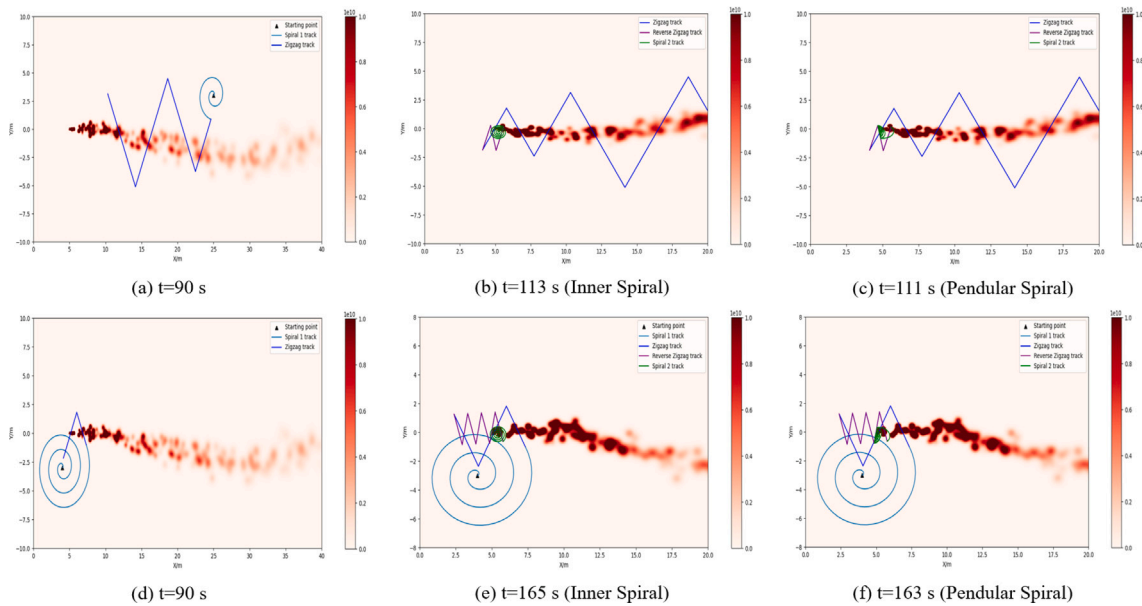


Fig. 13. Simulation results of the Zigzag-Spiral Algorithm. Figures (a), (b) present the results for starting points E of the Inner Spiral algorithm and Figures (a), (c) present the results of the Pendular Spiral algorithm. The blue line segment represents the Zigzag path, the purple line segment indicates the Reverse Zigzag path, and the brown line segment represents the novel spiral trajectory. Figures (d), (e) and Figures (d), (f) present the results for starting points F of the two algorithms.

conditions, a synchronized motion control strategy was implemented for the operation of robotic platforms. The experimental protocol was

structured in an iterative cycle, articulated as “concentration sampling → motion control → concentration sampling”, and was executed with

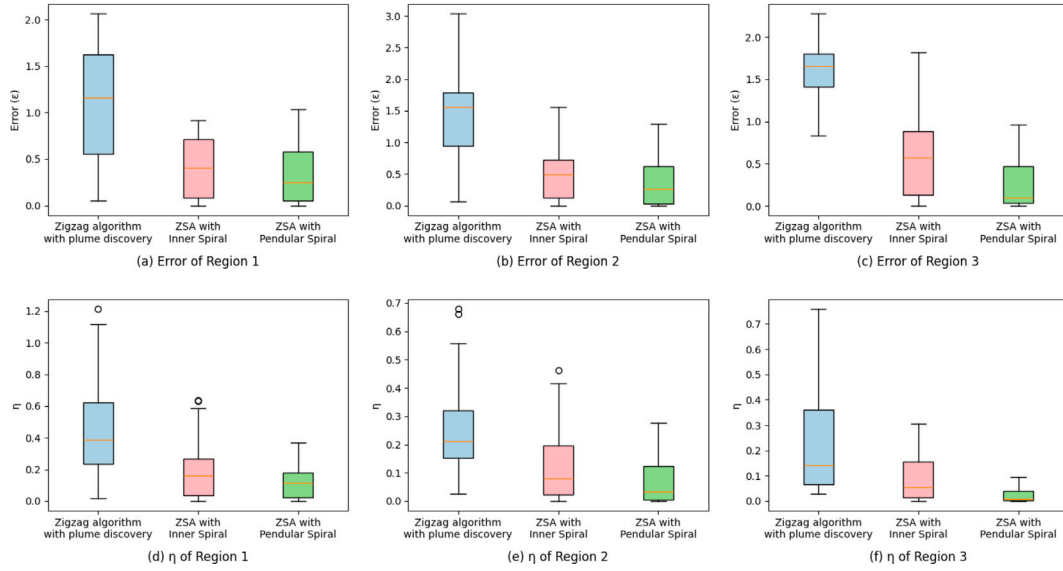


Fig. 14. Experimental results depicted as a box plot in the dynamic plume model.

Table 8

Experimental results in the dynamic plume model.

Method	Starting point	Error(m)	$\eta(10^{-3})$
Zigzag algorithm	E	1.612	79.692
	F	N/A	
ZSA with Inner Spiral	E	0.037	1.865
	F	0.043	13.431
ZSA with Pendular Spiral	E	0.027	1.359
	F	0.025	7.735

strict timing constraints. Each cycle, consisting of a sensory sampling phase and a motion execution phase, sustained fixed durations of 0.5 s for each phase, culminating in a total cycle period of 1.0 s.

In this section, the starting points were designated as $E = [25, 3]^T$ and $F = [4, 3]^T$. Following the procedure outlined in Section 3, the termination condition parameter and the concentration boundary threshold were established as $q = 0.1$ and $\tau = 1$, respectively. The path length thresholds derived from a large angle fold were set at $\epsilon = 0.6$ and $\epsilon' = 0.3$. The settings for the Zigzag Search are detailed in Table 7. Additionally, the parameters for the Spiral are set as follows: $b = 0.2$, $b_1 = 0.02$, $b_2 = \log(i + 1)$ (where i represents the line segment number as illustrated in Fig. 5), and $\gamma = 10$. The values of these parameters are also set according to the guidelines in Section 3.3, which are based on the relationship with the shorter side length of the search area. For the Gaussian diffusion model adopted in this paper, the shorter side length of the exploration area c is taken as 20 m. Consequently, the basic step length d_1 lies in the interval: $d_1 \in [0.1\sqrt{20}, 0.3\sqrt{20}] \approx [0.447, 1.342]$, and the basic spiral coefficient b satisfies $b = \frac{c}{4\pi n}$ with $n \geq 3$. For $c = 20$ this yields $0 < b \leq \frac{20}{12\pi} \approx 0.530$.

Given the time-varying nature of the system, the arc length for the Spiral path was fixed at 2 m for each second iteration. As illustrated in Fig. 13, which depict the complete motion trajectories of the robot initiating from a downwind position and a source-backside position respectively, the proposed ZSA effectively accomplishes the source-seeking process despite variations in concentration due to changes in puff size and position.

Experimental results under the dynamic model are summarized in Table 8. Both the error and η metrics reveal that ZSA, augmented by the fine search stage, attains lower positioning errors and enhanced search performance.

Given the inherent limitations of the Zigzag algorithm, a randomized starting position, whether determined through a simple random

walk or otherwise, does not consistently guarantee that the robot will commence its operation within the plume. To ensure a fair comparison and to validate the experimental evaluation criteria delineated in Section 2, comparative experiments were conducted. These experiments compared the basic Zigzag algorithm, enhanced with plume discovery capabilities, against variants of ZSA that incorporate Inner Spiral and Pendular Spiral strategies across various region sizes. The regions were defined as follows:

- Region 1: $\{(x, y) \mid x \in [0, 10], y \in [-2, 2]\}$,
- Region 2: $\{(x, y) \mid x \in [0, 20], y \in [-3, 3]\}$,
- Region 3: $\{(x, y) \mid x \in [0, 40], y \in [-5, 5]\}$.

Each region involved 100 experimental runs. A box plot of the results is presented in Fig. 14. The reduced number of outliers in this plot underscores the robustness of the proposed plume discovery strategy, which remains effective even under dynamic and turbulent conditions. Notably, as the search area expands, the positioning error tends to increase. Conversely, the corresponding η values decrease, reflecting an increase in the absolute path length required in larger regions. This observation emphasizes the utility of η as a comparative performance metric across different scales of regions. Table 9 reports the mean error and the mean η values, further demonstrating the enhancements in positioning accuracy achieved by ZSA.

4.3. Comparison with model-based algorithm

To provide a more direct and discriminative comparison, we evaluate our method against a model-based algorithm. As a typical algorithm in this category, we implement a particle filter-based source term estimation method, which follows the information-driven search framework introduced by Hutchinson et al. [27]. This algorithm serves as a suitable benchmark, since its action selection strategy is conceptually analogous to reinforcement learning methods, which are inherently model-based and probabilistic, thus enabling a meaningful comparison.

The robot's movement in the particle filter algorithm is guided by an information-based reward function. At each time step, the robot evaluates candidate actions from the admissible action set $\Psi = \{\uparrow, \downarrow, \leftarrow, \rightarrow\}$, corresponding to moving one step in each cardinal direction, and selects the one that maximizes the expected information gain. The expected utility of an action $a_k \in \Psi$ is approximated by generating hypothetical future measurements from the current particle set. The specific implementation details of this algorithm can be found at <https://github.com/Cunjia-Liu/AutoSTE>.

Table 9
Experimental results for different regions in the dynamic plume model.

Method	Region	Mean error $\bar{\rho}$	Mean ratio $\bar{\eta}$
Zigzag algorithm with plume discovery	Region 1	1.089	0.437
	Region 2	1.391	0.253
	Region 3	1.593	0.240
ZSA with Inner Spiral	Region 1	0.420	0.177
	Region 2	0.460	0.119
	Region 3	0.619	0.098
ZSA with Pendular Spiral	Region 1	0.338	0.117
	Region 2	0.354	0.074
	Region 3	0.265	0.037

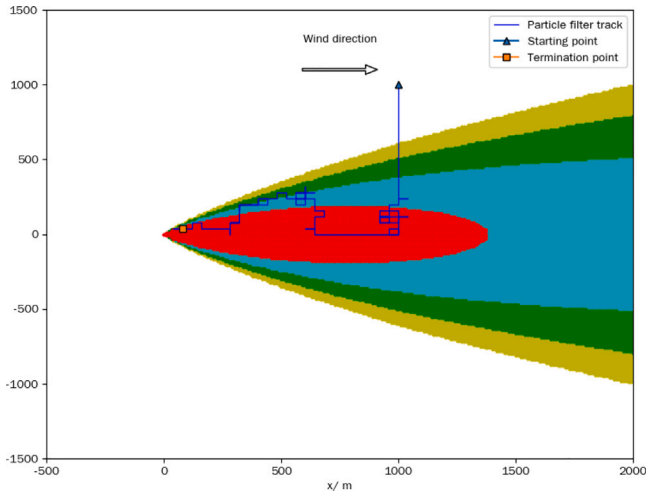


Fig. 15. Simulation results of the particle filter algorithm. Consistent with the previous text, the figure shows the specific movement path of the robot.

Table 10
Performance comparison of ZSA and particle filter algorithms.

Method	Mean error (m)	Mean time (s)
ZSA with Inner Spiral	6.19	2094.08
ZSA with Pendular Spiral	5.07	1845.44
Particle Filter ($N = 3000$)	5.94	2686.27
Particle Filter ($N = 6000$)	5.22	3749.21
Particle Filter ($N = 10,000$)	3.45	4921.95

The particle filter algorithm requires a large number of particles and necessitates real-time computation of concentration distributions for each hypothetical source relative to the current robot position, resulting in high computational load. We attempted to conduct experiments using the filament-based model mentioned above, but found that this model presents computational challenges in simulation implementation: not only the filament-based model requires a warm-up period for plume formation, but the particle filter algorithm also relies heavily on a large number of particles to ensure result accuracy. Moreover, the algorithm in [27] has been validated through incense stick combustion experiments, demonstrating its robustness for practical applications. Therefore, we ultimately adopted the Gaussian plume model described in Section 4.1 for algorithm implementation and performance evaluation.

Both algorithms were evaluated with a robot moving speed of 5 m/s. For the particle filter algorithm, the step size for the action set was set to 20 m. A total of 50 independent experiments were conducted for each algorithm.

As illustrated in Fig. 15, the trajectory of the robot under the particle filter algorithm is depicted. Table 10 summarizes the performance of ZSA and the particle filter algorithm with varying numbers of particles, comparing their mean localization errors and mean computation times.

The reported computation times in the results represent the overall experimental duration, which includes both the robot's movement time and the algorithm's inference time. The experimental results indicate that with a small number of particles ($N = 3000$ and $N = 6000$), the proposed ZSA with Pendular Spiral offers distinct advantages, achieving greater localization accuracy without introducing computational burden. When using a large number of particles ($N = 10,000$), the particle filter attains even higher accuracy, but this improvement comes at the cost of increased computation time. This performance difference arises from the particle filter's inherent dependence on prior knowledge of a dispersion model for likelihood evaluation. In contrast, ZSA operates without such model priors, thereby maintaining low computational cost while achieving an effective balance between localization accuracy and computational efficiency.

5. Physical experiments

5.1. Experimental setup

In addition to simulation-based validation, the proposed ZSA was experimentally evaluated in a ventilated, obstacle-free basement environment. The test area was a flat surface measuring (5×4) m. The hardware platform included an MQ-2 gas sensor (for data acquisition), a Khepera IV wheeled mobile robot, a gas source, and auxiliary modules such as a Wi-Fi router, a fan, and a camera. The MQ-2 sensor was integrated with the Khepera IV robot via external extension boards J700 and J701. Gas concentration data were transmitted through a UART serial interface, enabling the construction of an electronic nose system.

The gas source comprised an incense stick burning within a (5×5) cm container. Fig. 16 depicts the experimental setup, which includes both the hardware configuration and the layout of the testing environment. The role of the PC was confined to issuing commands for algorithm execution and collecting data, such as gas concentration and robot trajectory; it did not interfere with the process of motion control. The Khepera IV robot navigated autonomously, relying exclusively on real-time gas concentration measurements and the algorithm proposed, which dynamically computed the next target position to guide the robot.

Its sensitivity characteristics follow an empirical power-law relationship between the sensor resistance ratio (R_s/R_0) and gas concentration, where R_s is the sensor resistance in the target gas and R_0 is its resistance in clean air. The specific calibration used in our system is derived from the manufacturer's data and implemented internally by the sensor module's microcontroller as:

$$\text{ppm} = \left(\frac{11.5428 \times 35.904 \times V_{r_i}}{25.5 - 5.1 \times V_{r_i}} \right)^{0.6549} \quad (27)$$

where V_{r_i} is the analog output voltage of the sensor. The microcontroller performs this calculation and outputs the pre-calculated ppm value via the serial interface.

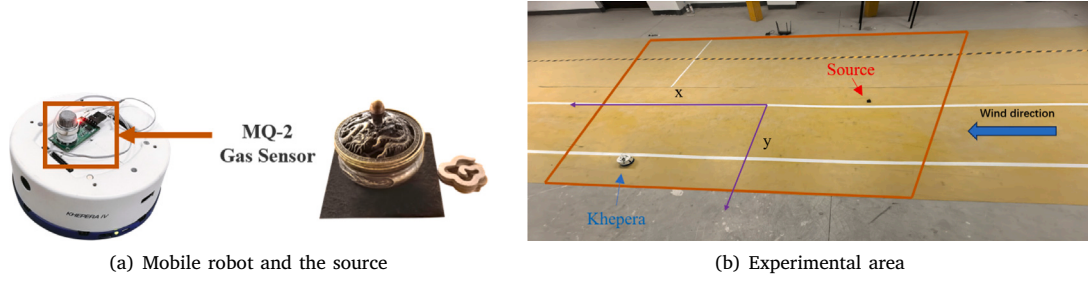


Fig. 16. Schematic diagram of the experimental setup: (a) shows the Khepera IV mobile robot and the gas source; (b) outlines the experimental area with a red bounding box, also defining the Cartesian coordinate system and the wind direction (pointing toward the positive x-axis).

Table 11

Summary of sensor time response parameters.

Parameter	Mean	Range
Response time, $T_{50, \text{response}}$ (s)	2.42	2.15–3.05
Response time, $T_{90, \text{response}}$ (s)	6.86	5.05–8.20
Recovery time, $T_{20, \text{recovery}}$ (s)	12.80	10.50–13.20

The data is transmitted in five-byte frames structured as: $[0xFF, 0xAA, B_3, B_4, 0x30]$, where B_3 and B_4 represent the high and low bytes, respectively. The gas concentration C (in ppm) is then calculated using:

$$C = (B_3 \times 256) + B_4. \quad (28)$$

The sensor module supports measurements ranging from 150 ppm to 10,000 ppm.

To establish a reliable detection threshold, we measured the sensor's baseline output in clean air over a 5-minute period. The measured baseline concentration followed a normal distribution with mean $\mu_0 = 332.33$ ppm and standard deviation $\sigma_0 = 18.51$ ppm. Based on this characterization, we set the plume detection threshold $\tau = \mu_0 + 3\sigma_0 = 387.86$ ppm, rounded to 400 ppm. Based on the three-sigma rule for normal distributions, this detection threshold ensures that the theoretical probability of a false positive due to baseline noise is approximately 0.135%.

To characterize the sensor's dynamic response in the presence of a gas source, we conducted repeated exposure tests. The robot equipped with the sensor was initially placed in a clean air region away from the gas source for 10 s to establish the baseline concentration C_0 . The robot was then rapidly moved to a position 50 cm downwind from the gas source (at (3.0, 2.0) m, with the source at (2.5, 2.0) m) and remained there for 30 s. Finally, the robot was quickly moved back to the clean air region to observe the recovery process.

Fig. 17 shows a representative response curve from this test procedure. Let t_0 denote the time when the robot begins moving toward the plume (response start), t_p the time when the concentration reaches its peak value C_p , and t_r the time when the robot begins moving away from the plume (recovery start). The following response time parameters are defined: (1) $T_{50, \text{response}}$: Time from t_0 until the concentration reaches $C_0 + 0.5(C_p - C_0)$; (2) $T_{90, \text{response}}$: Time from t_0 until the concentration reaches $C_0 + 0.9(C_p - C_0)$; (3) $T_{20, \text{recovery}}$: Time from t_r until the concentration falls to $C_p - 0.2(C_p - C_0)$. We repeated this experiment 10 times and calculated the statistical parameters shown in Table 11.

Given the highly dynamic and unpredictable nature of gas diffusion in real-world environments, maintaining high-frequency bidirectional communication for both real-time transmission of gas concentration data and reception of control commands poses a significant challenge. To ensure reliable sensor responsiveness, a four-second pause was introduced prior to each movement to allow adequate sensing. The selection of this pause duration is justified as follows: (1) For plume entry detection, the current threshold $\tau = 400$ ppm is not significantly lower than T_{50} (2.42 s), but it remains below T_{90} (6.82 s). Given the

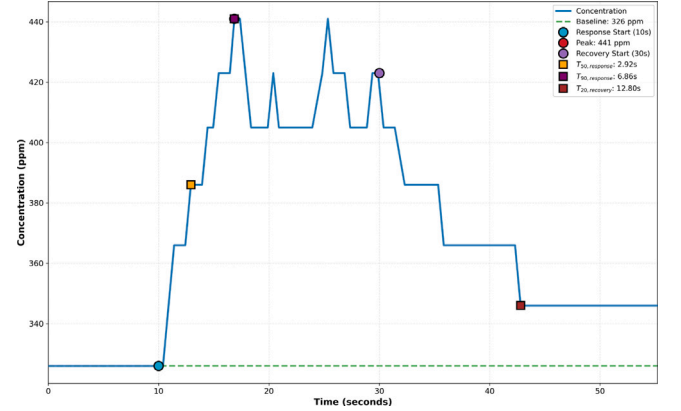


Fig. 17. Sensor response measurement result. The blue line represents the measured concentration, the green dashed line indicates the baseline C_0 , and the orange dashed line shows the peak level C_p . The cyan, red, and purple circles mark the response start t_0 , peak t_p , and recovery start t_r points, respectively. Orange, purple, and brown squares denote the $T_{50, \text{response}}$, $T_{90, \text{response}}$, and $T_{20, \text{recovery}}$ time markers.

actual pause duration of 4s and accounting for engineering delays (on the order of tenths of a second), the sensor still has sufficient time to reach a concentration well above τ . Thus, the system's response to plume entry is reliable and meets the requirements. (2) For plume exit detection, as noted previously, the slower recovery may lead to delayed recognition of plume exit. However, the algorithm incorporates robustness measures: the re-entry steps d_3 and d_6 , along with appropriately reduced distance thresholds ϵ and ϵ' (introduced in Section 3), provide sufficient tolerance to accommodate such sensing delays.

To ensure consistent gas distribution across multiple experimental trials and enable meaningful comparison of algorithm performance, the following measures were implemented:

- The basement was ventilated and then sealed for at least 30 min prior to each subsequent session, making sensors to return to baseline. All doors and windows remained closed during testing to minimize external airflow disturbances.
- Before each experimental run, the incense burned for 5 min to ensure a stable plume had developed. The source container was positioned at a fixed coordinate (2.5, 2.0) m within the defined coordinate system.
- A single fan was positioned at the negative x-boundary of the test area (outside the area to avoid interfering with robot movement) and operated at the same speed for every experiment. This established a consistent, unidirectional airflow along the positive x-axis to create a reproducible and standardized advection-dominated transport regime.

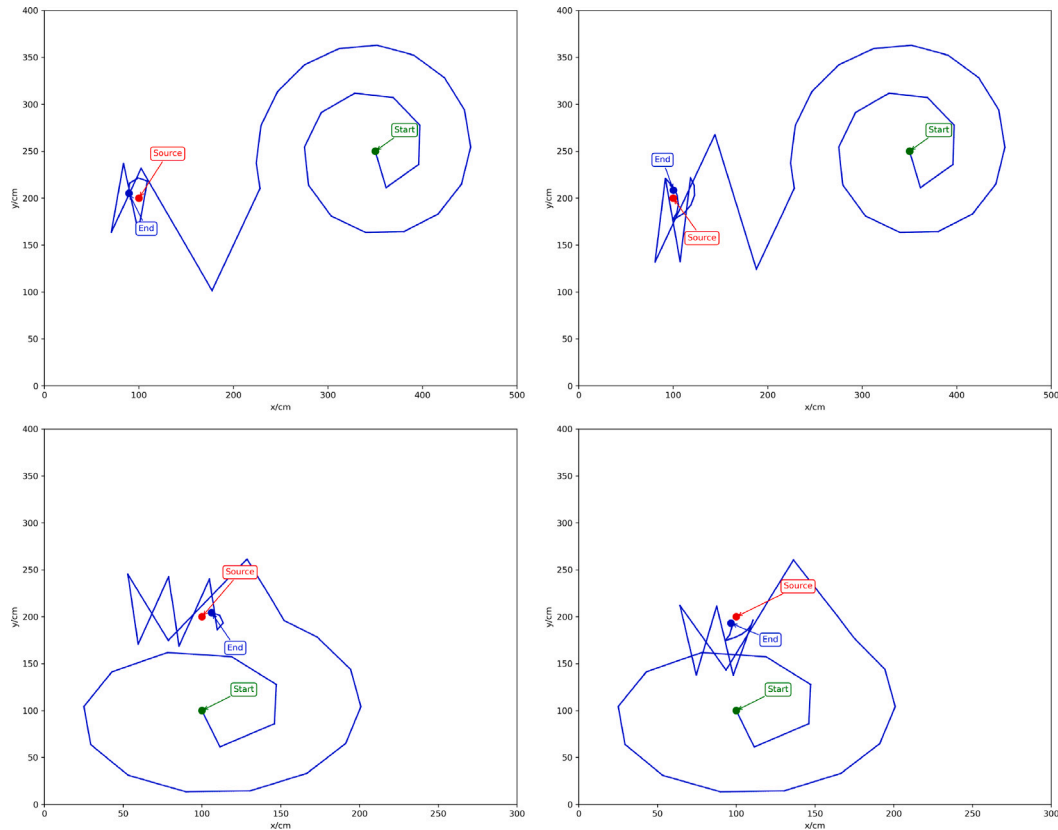


Fig. 18. Four representative results of the mobile robot's trajectory, obtained under scenarios where the initial position is located downwind and behind the source, using ZSA with Inner Spiral and ZSA with Pendular Spiral, respectively.

These controlled conditions ensured that the statistical properties of the gas plume remained consistent across trials, allowing us to attribute performance differences primarily to the algorithmic strategies rather than environmental variations. Detailed implementation and parameter configurations are documented at: <https://github.com/ustbxt/robot-source-seeking-git>.

5.2. Experimental result

Each algorithm underwent testing in ten trials: eight with the robot initially positioned downwind and two with it starting directly behind the source. Fig. 18 depicts the actual motion trajectories of the robot using ZSA with Inner Spiral and ZSA with Pendular Spiral enhancements. Due to the influence of natural airflow, plume diffusion was not consistently aligned with the positive x -axis, which resulted in variations in the points at which the robot entered or exited the plume across trials. Notably, a considerable portion of the path was dedicated to the discovery of the plume. This observation can be attributed to the limited sensing range of the MQ-2 sensor and the constrained size of the plume generated by the burning incense. Detectable concentrations were only observed in close proximity to the source, underscoring the necessity of effective plume localization strategies. Additionally, the four trajectories displayed did not fully execute the algorithm, as the next target point calculated by the planning was already very close to the source, with a distance of less than 10 cm. Consequently, an early stop command was implemented.

Table 12 summarizes the positioning error and execution time for each method across ten trials. The results indicate that the proposed ZSA achieves enhanced positioning accuracy, particularly when combined with the Pendular Spiral strategy. This improvement, however, is accompanied by a substantial increase in the average search duration—from approximately 131 s for the baseline Zigzag algorithm to over

204 s for ZSA with Pendular Spiral. For battery-powered autonomous platforms, operational duration is a primary determinant of mission endurance. While the present study did not include direct measurement of energy consumption, the extended runtime required by the Pendular Spiral strategy signifies a greater draw on the robot's energy budget, as all subsystems (motors, sensors) remain active for a longer period. This establishes a clear trade-off between localization precision and operational efficiency. The choice of strategy within the ZSA framework should therefore be guided by mission requirements. ZSA is the solution for missions where precision is the paramount, non-negotiable objective. This applies to high-stakes scenarios such as locating the epicenter of a hazardous chemical or gas leak for containment, pinpointing the source of an explosive or toxic substance for safe disposal, or accurately mapping the origin of a biological or environmental marker for scientific study. In these cases, the cost of additional search time and energy is justified by the critical need to minimize positioning error. Conversely, when the primary constraints are operational speed or limited energy reserves — such as in rapid, large-area environmental monitoring surveys, time-sensitive search missions, or operations with restricted battery capacity or recharge opportunities — a more efficiency-focused approach may be preferred.

It is also critical to note that effective gas sampling by the robot requires proper spatial alignment with the source. In this setup, the gas source and the reference wind direction were aligned on the horizontal plane at approximately the same height as the robot's sensor module. This limitation suggests that the current implementation is applicable solely to two-dimensional search tasks. Extending to three-dimensional scenarios is a significant direction for future research.

6. Conclusion

This study introduces a novel model-free ZSA for autonomous GSL using a mobile robot. The proposed algorithm incorporates a coarse-to-fine search mechanism, enabling the robot to transition from initial

Table 12

Experimental results in the actual scenario.

Method	Average error(cm)	Average time(s)
Zigzag algorithm with plume discovery	25.24	131.4
ZSA with Inner Spiral	11.19	218.8
ZSA with Pendular Spiral	10.08	204.5

plume detection to precise source localization without the reliance on environmental models or gradient information. Extensive experiments in both simulated and real-world environments have demonstrated the robust performance of ZSA under diverse initial conditions, including scenarios where the robot starts from a position behind the gas source. These results affirm the algorithm's adaptability and reliability in dynamic and uncertain environments.

Future work will focus on extending the proposed method to three-dimensional GSL. Furthermore, efforts will be directed towards enhancing algorithmic efficiency, reducing search time, and increasing robustness against environmental disturbances, with the aim of enabling practical deployment in real-world applications such as hazardous leak detection, environmental monitoring, and search-rescue operations.

CRediT authorship contribution statement

Tian Xu: Writing – review & editing, Writing – original draft, Methodology. **Xinmiao Sun:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Chuangdong Wu:** Writing – review & editing, Methodology. **Dawei Ding:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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